# Linear Algebra I — Week 4

## Row Reduction, Solving Linear Systems & Bases from RREF

Name: _____ Date: _____ All work **by hand** — no calculator, no software.

### Key Facts

**Elementary row operations.** (i) swap two rows $R_i \leftrightarrow R_j$; (ii) scale a row $R_i \to cR_i$ with $c \neq 0$; (iii) add a multiple of one row to another $R_i \to R_i + \lambda R_j$. They do not change the solution set of a system, and they can be undone, so they do not change the span of the rows.

**REF and RREF.** A matrix is in *row echelon form* (REF) if the zero rows are at the bottom and the first nonzero entry (the *leading entry*) of each row lies strictly to the right of the leading entry of the row above. It is in *reduced* row echelon form (RREF) if moreover every leading entry is 1 and is the only nonzero entry in its column. The RREF of a matrix is unique. A *pivot column* is a column containing a leading entry.

**Solving $A\mathbf{x} = \mathbf{b}$.** Reduce $\big(A \mid \mathbf{b}\big)$ to RREF. If a row reads $(0 \cdots 0 \mid 1)$ the system is *inconsistent*. Otherwise variables in pivot columns are *basic*, the others are *free*. No free variable: a unique solution. At least one free variable: infinitely many solutions, written as $\mathbf{p} + s\,\mathbf{v}_1 + t\,\mathbf{v}_2 + \cdots$ (parametric vector form).

**Homogeneous systems.** $A\mathbf{x} = \mathbf{0}$ is always consistent ($\mathbf{x} = \mathbf{0}$). Its solution set is a subspace, the span of the vectors multiplying the free parameters. If there are more unknowns than equations there is a free variable, hence a nonzero solution. If $\mathbf{p}$ solves $A\mathbf{x} = \mathbf{b}$, every solution is $\mathbf{p} + \mathbf{h}$ with $A\mathbf{h} = \mathbf{0}$.

**Columns, rows and RREF.** The columns of $A$ at the pivot positions of its RREF form a basis of $\mathrm{col}(A)$ (take them from the original $A$, not from the RREF). The nonzero rows of the RREF form a basis of $\mathrm{row}(A)$. The entries of a non-pivot column of the RREF are the coefficients expressing that column of $A$ through the pivot columns. Vectors $v_1, \ldots, v_k$ are independent exactly when the matrix with these columns has a pivot in every column.

## Worked Examples

### Example 1 · Reducing a matrix to RREF

Reduce $\begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 3 & 4 \end{pmatrix}$ to RREF.

**Solution.**

1. Go column by column. Column 1: the top entry is already 1, so use it as the pivot and clear the entries below.
2. Apply $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1$:

   $$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 3 \end{pmatrix}$$
3. Apply $R_1 \to R_1 - 2R_2,\ R_3 \to R_3 - R_2$:

   $$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$
4. Apply $R_3 \to \frac{1}{2}R_3$:

   $$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$
5. Apply $R_1 \to R_1 + R_3,\ R_2 \to R_2 - R_3$:

   $$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$
6. Every column has a pivot; the result is the identity matrix, which is in RREF.

**Answer:** RREF $= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

### Example 2 · A system with a unique solution

Solve $\begin{cases} x + y = 3 \\ 2x - y = 0 \end{cases}$

**Solution.**

1. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

   $$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & -1 & 0 \end{array}\right)$$
2. Apply $R_2 \to R_2 - 2R_1$:

   $$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -3 & -6 \end{array}\right)$$

3. Apply $R_2 \to \dfrac{1}{-3}R_2$:

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 2 \end{array}\right)$$

4. Apply $R_1 \to R_1 - R_2$:

$$\left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right)$$

5. Every column of the coefficient part holds a pivot, so there are no free variables and the solution is unique.

6. Read each nonzero row as an equation for its basic variable:

$x = 1$

$y = 2$

7. Check by substituting $(x, y) = (1, 2)$ into the original equations: the left sides give $(3, 0)$, the right-hand sides ✓.

**Answer:** Unique solution: $(x, y) = (1, 2)$.

## Example 3 · A system with infinitely many solutions

Solve $\begin{cases} x + y + z = 3 \\ 2x + y = 3 \end{cases}$

**Solution.**

1. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 1 & 0 & 3 \end{array}\right)$$

2. Apply $R_2 \to R_2 - 2R_1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & -1 & -2 & -3 \end{array}\right)$$

3. Apply $R_2 \to \dfrac{1}{-1}R_2$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 2 & 3 \end{array}\right)$$

4. Apply $R_1 \to R_1 - R_2$:

$$\left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 3 \end{array}\right)$$

5. Pivot columns are the columns of the basic variables: $x$, $y$. The remaining variable(s) $z$ are *free*: call them $z = s$.

6. Read each nonzero row as an equation for its basic variable:

$x = s$

$y = 3 - 2s$

$z = s$

7. Write this as a vector: $\mathbf{x} = \begin{pmatrix}0\\3\\0\end{pmatrix} + s\begin{pmatrix}1\\-2\\1\end{pmatrix}$ (the parameters $s$ may be any numbers).

8. Check with $s = 1$: $\mathbf{x} = \begin{pmatrix}1\\1\\1\end{pmatrix}$, and substituting into the original equations gives the right-hand sides $(3, 3)$ ✓.

**Answer:** Infinitely many solutions: $\mathbf{x} = \begin{pmatrix}0\\3\\0\end{pmatrix} + s\begin{pmatrix}1\\-2\\1\end{pmatrix}$.

## Example 4 · An inconsistent system

Solve $\begin{cases} x + y + 2z = 1 \\ 2x + 2y + 5z = 3 \\ x + y = 2 \end{cases}$

**Solution.**

1. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 2 & 2 & 5 & 3 \\ 1 & 1 & 0 & 2 \end{array}\right)$$

2. Apply $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -2 & 1 \end{array}\right)$$

3. Apply $R_1 \to R_1 - 2R_2,\ R_3 \to R_3 + 2R_2$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 \end{array}\right)$$

4. Row 3 reads $0 = 1$, a contradiction. No numbers $x, y, z$ can satisfy it, so the system is *inconsistent*.

**Answer:** No solution (inconsistent).

## Example 5 · A homogeneous system

Solve $\begin{cases} x + y + 2w = 0 \\ 2x + 2y + z + 5w = 0 \end{cases}$ and write the solution set as a span.

**Solution.**

1. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 2 & 0 \\ 2 & 2 & 1 & 5 & 0 \end{array}\right)$$

2. Apply $R_2 \to R_2 - 2R_1$:

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{array}\right)$$

3. Pivot columns are the columns of the basic variables: $x$, $z$. The remaining variable(s) $y$, $w$ are *free*: call them $y = s$, $w = t$.
4. Read each nonzero row as an equation for its basic variable:

   $x = -s - 2t$

   $y = s$

   $z = -t$

   $w = t$

5. Write this as a vector: $\mathbf{x} = s\begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t\begin{pmatrix} -2 \\ 0 \\ -1 \\ 1 \end{pmatrix}$ (the parameters $s$, $t$ may be any numbers).
6. Check with $s = 1$, $t = 1$: $\mathbf{x} = \begin{pmatrix} -3 \\ 1 \\ -1 \\ 1 \end{pmatrix}$, and substituting into the original equations gives the right-hand sides $(0, 0)$ ✓.

**Answer:** Infinitely many solutions: $\mathbf{x} = s\begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t\begin{pmatrix} -2 \\ 0 \\ -1 \\ 1 \end{pmatrix}$. The solution set is span of the two vectors above.

## Example 6 · Column space basis and column relations

For $A = \begin{pmatrix} 1 & 3 & 0 & 2 \\ 2 & 6 & 1 & 5 \\ 1 & 3 & 1 & 3 \end{pmatrix}$ find a basis of $\mathrm{col}(A)$ and express the other columns using it.

**Solution.**

1. Reduce $A$ to RREF.
2. Apply $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - R_1$:

   $$\begin{pmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

3. Apply $R_3 \to R_3 - R_2$:

   $$\begin{pmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

4. Pivot columns: 1,3. By the pivot-column rule, a basis of $\mathrm{col}(A)$ is formed by columns 1,3 *of the original* $A$: $\left\{\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}\right\}$.
5. The non-pivot columns of the RREF give the coefficients: column 2 of $R$ is $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$, so $\mathrm{col}_2 =$

$3\,\mathrm{col}_1$; column 4 of $R$ is $\begin{pmatrix}2\\1\end{pmatrix}$, so $\mathrm{col}_4 = 2\,\mathrm{col}_1 + 1\,\mathrm{col}_3$.

**Answer:** Basis $\{\begin{pmatrix}1\\2\\1\end{pmatrix}, \begin{pmatrix}0\\1\\1\end{pmatrix}\}$.

## Example 7 · A system over $\mathbb{C}$

Solve $\begin{cases} x + iy = 1 + i \\ ix + 2y = i \end{cases}$ over $\mathbb{C}$.

**Solution.**

1. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{cc|c} 1 & i & 1+i \\ i & 2 & i \end{array}\right)$$

2. Apply $R_2 \to R_2 - iR_1$:

$$\left(\begin{array}{cc|c} 1 & i & 1+i \\ 0 & 3 & 1 \end{array}\right)$$

3. Apply $R_2 \to \frac{1}{3}R_2$:

$$\left(\begin{array}{cc|c} 1 & i & 1+i \\ 0 & 1 & \frac{1}{3} \end{array}\right)$$

4. Apply $R_1 \to R_1 - iR_2$:

$$\left(\begin{array}{cc|c} 1 & 0 & 1+\frac{2}{3}i \\ 0 & 1 & \frac{1}{3} \end{array}\right)$$

5. Every column of the coefficient part holds a pivot, so there are no free variables and the solution is unique.

6. Read each nonzero row as an equation for its basic variable:

$$x = 1 + \frac{2}{3}i$$
$$y = \frac{1}{3}$$

7. Check by substituting $(x, y) = (1 + \frac{2}{3}i, \frac{1}{3})$ into the original equations: the left sides give $(1 + i, i)$, the right-hand sides ✓.

**Answer:** Unique solution: $(x, y) = (1 + \frac{2}{3}i, \frac{1}{3})$.

## Example 8 · Proof: row reduction does not change the solutions

Prove: swapping two equations, scaling an equation by $c \neq 0$, or adding a multiple of one equation to another does not change the solution set.

**Solution.**

1. Swapping: the system is the same list of equations in a different order, and a vector solves a

list of equations regardless of order.

2. Scaling $E_i$ by $c \neq 0$: if $E_i(\mathbf{x}) = \beta_i$ then $(cE_i)(\mathbf{x}) = c\beta_i$. Conversely, if $(cE_i)(\mathbf{x}) = c\beta_i$, multiply both sides by $c^{-1}$ (possible because $c \neq 0$ in a field) to recover $E_i(\mathbf{x}) = \beta_i$.
3. Adding $\lambda E_j$ to $E_i$: if $\mathbf{x}$ solves both $E_i, E_j$ it solves $E_i + \lambda E_j$. Conversely, if $\mathbf{x}$ solves $E_j$ and $E_i + \lambda E_j$, then $E_i(\mathbf{x}) = (E_i + \lambda E_j)(\mathbf{x}) - \lambda E_j(\mathbf{x}) = \beta_i$.
4. Each operation is undone by an operation of the same kind, so solutions are preserved in both directions.

**Answer:** The three operations preserve the solution set, so a system and its RREF have the same solutions.

## Practice Problems

### Part A — Echelon Forms and Row Reduction

**1.** [Foundational]

Classify each matrix as: in RREF, in REF but not RREF, or neither. For every matrix that is not in RREF, name the condition it violates.

(a) $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ (c) $\begin{pmatrix} 0 & 1 & 3 \\ 1 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$ (d) $\begin{pmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

**2.** [Foundational]

Use elementary row operations to reduce $\begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 3 & 4 \end{pmatrix}$ to RREF. Write each operation in the form $R_i \to R_i - cR_j$ and state the pivot columns.

**3.** [Intermediate]

Reduce $\begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 0 & 2 \\ 1 & 2 & 1 & -1 \end{pmatrix}$ to RREF. State the pivot columns and explain why column 2 is not a pivot column.

**4.** [Intermediate]

Reduce $\begin{pmatrix} 0 & 2 & 4 \\ 1 & 1 & 1 \\ 2 & 0 & -2 \end{pmatrix}$ to RREF. A row swap is needed: say why, and state the pivot columns.

**5.** [Challenge]

Over $\mathbb{C}$, reduce $\begin{pmatrix} 1 & i & 0 \\ 2 & 2i & 1 \\ 1 & i & 1 \end{pmatrix}$ to RREF. State the pivot columns.

**6.** [Challenge]

(Why row reduction works.) Let $S'$ be obtained from a system $S$ by one elementary row operation $R_i \to R_i + \lambda R_j$ $(i \neq j)$. Prove that $S$ and $S'$ have the same solutions.

## Part B — Solving Linear Systems

**7.** [Foundational]

Solve by row reduction: $\begin{cases} x + y = 3 \\ 2x - y = 0 \end{cases}$

**8.** [Foundational]

Solve by row reduction: $\begin{cases} x + y + 2z = 5 \\ 2x - y + z = 4 \\ x + 2y - z = 3 \end{cases}$

**9.** [Intermediate]

Solve: $\begin{cases} x + 2y - z = 4 \\ 2x + 5y + z = 9 \end{cases}$ Describe the solution set in parametric vector form.

**10.** [Intermediate]

Decide whether the system is consistent: $\begin{cases} x + 2y + z = 1 \\ 2x + 4y + 3z = 3 \\ x + 2y - z = 2 \end{cases}$

**11.** [Intermediate]

Solve $\begin{cases} x + y + z + w = 2 \\ x + 2y + 3w = 5 \end{cases}$ and identify the basic and free variables.

**12.** [Intermediate]

Solve the homogeneous system $\begin{cases} x + 2y - w = 0 \\ 2x + 4y + z + w = 0 \\ x + 2y + z + 2w = 0 \end{cases}$ and write the solution set as a span.

**13.** [Intermediate]

Over $\mathbb{C}$, solve $\begin{cases} x + iy = 1 + i \\ ix + 2y = i \end{cases}$

**14.** [Challenge]

For which real $k$ does the system $\begin{cases} x + y + z = 1 \\ x + 2y + 3z = 2 \\ x + 3y + kz = 4 \end{cases}$ have (i) a unique solution, (ii) no solution, (iii) infinitely many solutions?

**15.** [Challenge]

A homogeneous system has $m$ equations and $n$ unknowns with $n > m$. Prove it has a nonzero solution.

## Part C — Column Spaces, Relations and Proofs

**16.** [Foundational]

The matrix $R = \begin{pmatrix} 1 & 0 & 3 & 0 & -2 \\ 0 & 1 & -1 & 0 & 4 \\ 0 & 0 & 0 & 1 & 5 \end{pmatrix}$ is the RREF of the coefficient matrix of a homogeneous system. State the pivot columns, the basic and free variables (use $x_1, \dots, x_5$), and the general solution of $R\mathbf{x} = \mathbf{0}$.

**17.** [Intermediate]

Let $A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 1 & 2 \end{pmatrix}$. (a) Reduce $A$ to RREF. (b) Give a basis of the column space $\mathrm{col}(A)$ made of columns of $A$ itself. (c) Express each non-pivot column of $A$ as a linear combination of the pivot columns.

**18.** [Intermediate]

For $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{pmatrix}$, reduce to RREF and give a basis of the row space $\mathrm{row}(A)$ and the number of pivots.

**19.** [Intermediate]
Decide whether $\mathbf{w} = (3, 2, 1)$ lies in $\text{span}\{(1, 0, 1), (2, 1, 0), (0, 1, -2)\} \subseteq \mathbb{R}^3$ by row reducing one augmented matrix. Give the coefficients if it does.

**20.** [Intermediate]
Place the vectors $(1, 2, 1, 0),\ (1, 0, 1, 1),\ (2, 1, 0, 1)$ of $\mathbb{R}^4$ as columns of a matrix. Use RREF to decide whether they are linearly independent.

**21.** [Intermediate]
Show that $(1, 0, 1), (1, 1, 0), (0, 1, 1), (2, 3, 1)$ in $\mathbb{R}^3$ are linearly dependent, and use RREF to find an explicit relation among them.

**22.** [Intermediate]
In $P_2(\mathbb{R})$, decide whether $q(x) = 2 + 3x + x^2 \in \text{span}\{1 + x,\ x + x^2,\ 1 + x^2\}$. Set up the system, reduce, and conclude.

**23.** [Challenge]

Let $A\mathbf{x} = \mathbf{b}$ be consistent and let $\mathbf{p}$ be one solution. Prove that the solution set is exactly $\{\mathbf{p}+\mathbf{h} : A\mathbf{h} = \mathbf{0}\}$.

**24.** [Challenge]

Find the condition on $a, b, c$ for the system $\begin{cases} x + 2y - z = a \\ 2x + 3y + z = b \\ x + y + 2z = c \end{cases}$ to be consistent.

**25.** [Challenge]

Let $R$ be the RREF of $A$. Prove that every non-pivot column of $R$ is a linear combination of the pivot columns of $R$, and explain why the same relation holds among the corresponding columns of $A$.